

ADVANTAGE Section Discussion



Drilling Fluids

Operator Peak Petroleum Indust. Limited	Field Bilabri	Block/Co/Parish OML - 122 / Nigeria	Facility Bilabri DX 1
Well Bilabri DX 1	Rig BULFORD DOLPHIN	Job	Bilabri DX 1

Section Details - 12-1/4" Section - #4

Mud Type Invert Emulsion	Mud System SYN-TEQ (PARA)	Top MD 1550.00 m	Bot MD 2779.00 m
Start-End Date 27/Dec/2005-12/Jan/2006	Open Hole Diam in	Total Days 16	Sect Length 1229.00 m
Section Comment Drill 12-1/4" hole w/SBM to section TD. Run & cmt 9-5/8" csg			

Section Discussion

A 12-1/4" slick assy was run in hole to tag cement at 1497mts. Cement plug & float collar was drilled to 1525m with 9.1 ppg water base mud from the 17-1/2" section. The cmt which was found to be soft below 1504m took only 0-2k to drill.

An initial Synthetic Oil Base Mud (SOBM) was delivered into isolated tanks, taking on board 540 bbls of 9.6 ppg instead of 10.0 ppg due to settling in the boat tanks. Mud was conditioned & weighted up to 10.ppg while waiting for the rest of the same. Hence, the continuation of cement drilling & float Collar was put on hold till second boat arrived with SOBM though with some delay for permission to discharge same from town having arrived location.

Once taken the required volume of SOBM. Continued drilling cement & shoe track with WBM from previous 17-1/2" TD. Of 1550 mts before drilling 3mts of new formation. Perform LOT subsequently with 9.1 ppg WBM resulting in 12.65 ppg. Equivalent Mud Density (E.M.D) Thereafter, hole was drilled further to 1575 mts for a second LOT to EMD = 12.45 ppg with 9.2 ppg WBM. Wellbore was subsequently displaced to 10.0 ppg SOBM by pumping weighted water base spacer, followed by base oil & then the SOBM.

Displacement stopped once the base oil spacer got to surface. While pulling out of the hole 12-1/4" slick assembly, sand traps & rest of WBM was dumped. Pits, surface lines & facilities were washed & flushed to avoid any further contamination. Also gates were closed & gelled up while the shakers were temporarily dressed up with 110 mesh screen sizes pending when mud sheared enough while circulating. Smaller screen sizes, were then accommodated later on, when Shale Shakers were able to handle a stronger pump rate.

A new 12-1/4" directional assembly with Auto-track tool (BHI) was run in the hole to bottom at 1575 mts, Toke surveyed & commenced drilling to 1583m. Observed problem with downhole tool. Tripped out for swapping tool.

During tripping, shakers screens were changed from 110 to another temporary sizes of 150s with, and also mud cleaner screens were changed from 175 to 210s.

Returned back in hole, with new Auto-track, and continued drilling ahead with a predominantly very fine sandy formation. Desander & desilter were run over the mud cleaner & centrifuges on desanding mode.

Continue to observe problems with Auto-track tool so another trip had to be made at 1600 mts. While tripping screens were further change to '200, 2x175' on the each Shaker to further screen down the mud as it shears and finer sand being drilled & more anticipated.

Continue to drill from 2014 mts 2216 mts. Losing mud at the shakes. Pump 50 bbls Hi-vis sweep, circulated. Continue drilling to 2240mts. F/F rlg leak at 12242-2250-2260 - 2270-2290. All ok

Continue drilling ahead problem free, observed considerable gas percentage during connection, circulated. Bottoms up, check for flow, and continue. drilling to 2477 mts Increase mud weight to 10.2 ppg Continue drilling ahead to 2560m and raise once more mud weight to 10.4 ppg. Circulated bottoms up and collect sample, continue to circulated. till clean returns. Boost riser through kill line, perform wiper trip from 2779 mts to 1525 mts without problems. Return to bottom, observe considerable gas. Increase Mud weigh to 10.5 ppg, circ & continue to observe hi level gas (21%) raise mud weight to 10.6 ppg. & continue circulating. Pull out to surface problem free rig up wire log and run in TTRM & DSL

Finished samples collection on log. No 4. Pulled out logging tool to covered samples. Rig down Baker Atlas, Make up slick BHA and trip in, @ 1500m Circulated bottoms up while conditioning Mud system. Continue tripping to 2000m. Circulated and condition Mud. Max Gas = 3100 Unit.

Continue to trip in. To TD @ 2779m. Circulated while conditioning mud and observed well behavior. Max gas @ 7010 Units. Continue to circulated and raise mud weight to 10.7 ppg. Check for flow. Pumped and spotted 50 bbls sweep on bottom. And pull out. To run 9 5/8" casing. Place casing and set the same, observe problem while setting new seal and become mechanically stuck due to low performance in the setting tool. Set free assembly and pull out problem free.